

Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_best_single_fixed_heuristic, phase8_full_solver_evolution, phase8_llm_evolved_adaptive_controller, phase8_random_portfolio.

Run Metadata

- run_name: run_20260520_175541_d
- started_at_local: 2026-05-20 17:55:41
- finished_at_local: 2026-05-20 18:25:17
- duration_hhmm: 00:30
- duration_seconds: 1775.739
- seed_offset: 3000
- replicate_label: d
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	1454.958643	0.07226	0.007284	0.03122	0.0	0.0	True
phase8_random_portfolio	random_portfolio	none	0.218775	0.146515	310.9475	0.218775	0.082909	0.132965	0.0	0.0	True
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	1558.234271	0.077389	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLI B Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	1493.702814	0.074184	0.007284	0.03122	0.0	0.0	False
phase8_llm_static_selector	static_selector	none	0.120261	0.048001	7173.831929	0.59296	0.001723	0.045395	0.0	0.62	False
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.214311	0.142051	383.137643	0.214311	0.023867	0.09403	0.0	0.62	True
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.214732	0.142472	545.303743	0.214732	0.027988	0.096788	0.548632	0.755	False
phase8_full_solver_evolution	full_solver	none	0.156335	0.084075	383.862343	0.156335	0.001014	0.058238	0.717957	0.76	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1454.958643 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.218775, selector regret 0.146515, runtime-adjusted gap 0.218775.
- Family transfer 0.082909 and combined transfer 0.132965.
- Runtime 310.9475 ms, runtime inflation -0.786284, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.077389.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 1558.234271 ms, runtime inflation 0.070982, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.074184.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1493.702814 ms, runtime inflation 0.026629, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.120261, selector regret 0.048001, runtime-adjusted gap 0.59296.
- Family transfer 0.001723 and combined transfer 0.045395.
- Runtime 7173.831929 ms, runtime inflation 3.930609, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:farthest_insertion_two_opt:edge_preserving_restart:stag3:cand2:restart1.
- Controller rule summary: {"acceptance_bias": 0.005, "candidate_limit_offset": 2, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic one-shot portfolio selector using instance-structure cues; base is farthest-insertion + 2-opt, with specialized fallbacks for clustered/bottleneck and edge-preserving recovery.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "farthest_insertion_two_opt", "low_improvement_threshold": 0.06, "name": "static_instance_adaptive_farthest_base", "nearest_neighbor_trap_heuristic": "cheapest_insertion_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 3, "temperature_scale": 1.05, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.023867 and combined transfer 0.09403.
- Runtime 383.137643 ms, runtime inflation -0.736668, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: nearest_neighbor_multistart:cheapest_insertion_two_opt:farthest_insertion_two_opt:edge_preserving_restart:stag3:cand3:restart1.
- Controller rule summary: {"acceptance_bias": 0.005, "candidate_limit_offset": 3, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "cluster_first_local_search", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selection among frozen TSP heuristics with online stagnation/restart tuning (interpretable structural triggers).", "failed_perturbation_threshold": 2, "grid_like_heuristic": "farthest_insertion_two_opt", "low_improvement_threshold": 0.08, "name": "det_adaptive_frozen_portfolio_v1", "nearest_neighbor_trap_heuristic": "edge_preserving_restart",

"random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 3, "temperature_scale": 1.1, "time_budget_trigger": 0.6, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.214732, selector regret 0.142472, runtime-adjusted gap 0.214732.
- Family transfer 0.027988 and combined transfer 0.096788.
- Runtime 545.303743 ms, runtime inflation -0.62521, mean novelty 0.548632, and complexity 0.755.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:farthest_insertion_two_opt:edge_preserving_restart:stag2:cand-1:restart2.
- Controller rule summary: {"acceptance_bias": 0.004, "candidate_limit_offset": -1, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive scheduler over frozen portfolio. Strong generalist baseline is farthest-insertion + 2-opt. Grid-like instances switch to candidate-pruned 2-opt. Strong nearest-neighbor trap risk uses multi-start NN (less thrashy than annealed).", "failed_perturbation_threshold": 1, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.09, "name": "det_adaptive_tsp_frozen_portfolio_epoch3", "nearest_neighbor_trap_heuristic": "nearest_neighbor_multistart", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 2, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 2, "temperature_scale": 1.1, "time_budget_trigger": 0.7, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.156335, selector regret 0.084075, runtime-adjusted gap 0.156335.
- Family transfer 0.001014 and combined transfer 0.058238.
- Runtime 383.862343 ms, runtime inflation -0.73617, mean novelty 0.717957, and complexity 0.76.
- Pareto efficient: True.

Judge Appendix

Conservative Interpretation of TSP Portfolio Suite Results

Metric	Notes / Priority
Primary: Held-out TSPLIB Gap	- Best performance by **phase8_best_single_fixed_heuristic** (gap 0.07226).
	- Several adaptive selectors (including **phase8_supervised_ml_selector**) match this gap.
	- Oracle selector (upper bound) also at 0.07226 gap but with higher runtime (not deployable).
Selector Regret vs Oracle	- Best fixed heuristic & supervised selector show zero selector regret (match oracle gap).
	- Adaptive controllers show substantial selector regret (~0.08–0.14), indicating imperfect adaptation.

Metric	Notes / Priority
Runtime-Adjusted Gap	- Best single fixed heuristic achieves 0.07226 gap with no runtime inflation.
	- Oracle's runtime-adjusted gap slightly higher (0.0774) with moderate (7%) runtime inflation.
	- Adaptive controllers have higher runtime-adjusted gaps (~0.15–0.21), but with reduced runtime inflation (mostly negative, i.e., faster).
	- Static LLM instance-adaptive selector suffers from large runtime inflation (~4x slower) and worsened TSPLIB gap (0.12).
Pareto Efficiency	- Pareto-efficient: best fixed heuristic, full solver evolution, evolved adaptive controller, and random portfolio.
	- Oracle and LLM static selector are <i>*not*</i> Pareto efficient due to runtime or gap trade-offs.
Cross-Family Transfer Gaps	- Best fixed heuristic and supervised selector show low transfer gaps (~0.03).
	- Adaptive controllers show higher transfer gaps (~0.06–0.09), reflecting some generalization challenges.
Interpretable Instance-Adaptive Control	- Adaptive controllers embody interpretable rules/triggers (e.g., stagnation thresholds, instance-structure cues).
	- However, they do not significantly close the gap to oracle or fixed heuristic baseline.
	- They offer runtime savings (negative runtime inflation) but at cost of increased final gap and regret.
Summary Points	- The strong baseline is a fixed heuristic: <i>*farthest_insertion_two_opt*</i> .
	- Learned/adaptive selectors approximate oracle TSPLIB gap without outperforming baseline.
	- Runtime overheads vary: some adaptive methods reduce runtime modestly but accept higher gaps.
	- LLM static selector is expensive runtime-wise and yields higher gaps; not Pareto optimal.
	- No novel heuristics discovered; control over known heuristics is consistent with design goals.

Final Conservative Conclusion

The best single fixed heuristic (farthest_insertion + 2-opt) remains a strong, Pareto-efficient baseline with the lowest held-out TSPLIB gap and zero selector regret. Adaptive/selective portfolio methods show interpretable instance-adaptive control, trading some accuracy for runtime efficiency, but do not surpass the fixed heuristic's gap or approach the oracle selector closely enough to justify runtime or gap costs. LLM-based static adaptation incurs excessive runtime inflation and worsened gaps. Overall, the results underscore the challenge of improving upon a robust fixed heuristic baseline via adaptive control, emphasizing careful trade-offs between solution quality, runtime, and generalization across TSP families without claiming new algorithmic improvements.